

Terravix — autonomous weeding for row crops

The Problem

Herbicide resistance is spreading and regulatory pressure on chemical weed control is increasing across Europe. Mechanical weeding is the alternative, and it is labour-intensive at a moment when farm labour is scarce and expensive.

Our Solution

Terravix is an autonomous field robot that identifies and removes weeds mechanically, without chemicals. Our computer vision model distinguishes crop from weed at 96.4% accuracy across eleven crop types, verified over two seasons of field trials.

The machine covers 1.4 hectares per hour, operates unsupervised for up to nine hours, and handles row spacing from 25 to 75 centimetres. Three generations of hardware have been built and field-tested.

This is genuinely difficult engineering and it works.

The Market

European arable land under row crops exceeds 40 million hectares. Regulatory phase-outs of several key herbicide actives are scheduled over the next six years.

Traction

We have completed two full seasons of field trials with six partner farms across France and Germany. The trials validated accuracy, throughput and reliability.

We do not yet have paying customers. The trials were conducted at our cost.

The Team

Élodie Marchand holds a doctorate in agricultural robotics and led the machine vision work. Jonas Brenner built the mechanical platform. Thirty-one people, twenty-four in engineering.

Business Model

The unit cost to manufacture is currently €118,000. We intend to sell at €140,000, or offer a seasonal service contract.

Farm economics at this price are difficult. A 200-hectare operation would need roughly nine years to recover the purchase against saved herbicide and labour cost. We are working to bring manufacturing cost down through volume and design simplification, and we believe a price point below €70,000 is achievable at scale.

We have not yet converted a trial farm into a purchase.

Funding

We are raising €15M to reduce unit cost, build a commercial team and convert the trial farms.